Real-Time Aerial Detection and Reasoning on Embedded-UAVs in Rural Environments

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***Abstract*— We present a unified pipeline architecture for a real-time detection system on an embedded system for unmanned aerial vehicles (UAVs). Neural architectures have been the industry standard for computer vision. However, most existing works focus solely on concatenating deeper layers to achieve higher accuracy with run-time performance as the tradeoff. This pipeline of networks can exploit the domain-specific knowledge on aerial pedestrian detection and activity recognition for the emerging UAV applications of autonomous surveying and activity reporting. In particular, our pipeline architectures operate in a time-sensitive manner, have high accuracy in detecting pedestrians from various aerial orientations, use a novel attention map for multiactivities recognition, and jointly refine its detection with temporal information. Numerically, we demonstrate our model's accuracy and fast inference speed on embedded systems. We empirically deployed our prototype hardware with full live feeds in a real-world open field.**

***Index Terms*— Deep learning in UAVs, human activity recognition, remote sensing, scene understanding.**

I. Introduction

REAL-TIME awareness and understanding of the physical world have long been open questions for the robotics community. For machines to successfully maneuver in the presence of dynamic obstacles, one must perceive the intention of the humans and respond appropriately. A real-time anomaly monitor system is crucial for crowd monitoring during mega-events [1], where the system must recognize any hostile intentions that any individuals might be planning to perform. Moreover, an autonomous aerial surveying system in a search-and-rescue mission [2] will require the ability to comprehend the state of any detected humans and reports accordingly [3], significantly when the number of dispatched drones greatly exceed the number of human operators [4].

The scope of this article is on performing decentralized real-time autonomous activity reporting in a remote area, focusing on semantic understanding of people's current status from the perspective of unmanned aerial vehicles (UAVs). UAVs play a crucial role when operating in rural environments, for tasks such as disaster response and recovery operations, where ground access is infeasible in the aftermath of natural disasters such as earthquakes. In disastrous situations, autonomous UAVs surveying is hugely desirable due to the area of land they can cover. Victims can be located faster and reached earlier when UAVs are deployed in mountain search-and-rescue operations [5]. However, current approaches transfer multigigabytes of video footage from remote drones to a centralized server for analysis. Network activities are severely limited in disaster area [6], and any wireless bandwidth should be preserved for the communication channel between emergency services [7] and critical communication. Therefore, it severely limits relying on centralized servers for video processing.

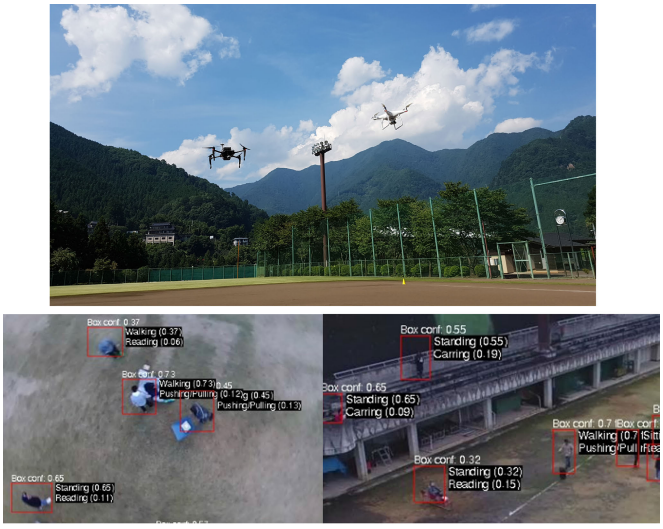


Fig. 1. (Top) Deploying live detection and recognition in the field of *Okutama, Japan*. (Bottom) Spatial locations of detected pedestrians, recognized actions, and confidence scores are all processed onboard. The processed information is overlaid on original video images for visualization.

On the other hand, lightweight models such as MobileNetV2 [8] can achieve reasonable runtime speed but sacrifice predictive performance on aerially captured images due to the highly varying camera angles from UAVs' perspective. Although they are suitable for static embedded systems, the severely limited inference quality poses difficulty for usage on UAVs. To this end, we propose our framework with the following properties. We will employ UAVs with an embedded camera on board to survey an area of interest and report any abnormality to operators, namely the current status of the detected pedestrians and associated GPS coordinates. The problem setup requires a sizeable amount of inexpensive UAVs patrolling the vast aerial space for search and rescue; therefore, each UAV must operate autonomously and raise alerts if necessary. Each UAV will be equipped with an embedded system and a pretrained model for surveying.

This article proposes a lightweight model pipeline that simultaneously detects pedestrians and recognizes their current status. Furthermore, our model is lightweight with minimal parameters and suitable for real-time processing in an onboard system. A novel regression-based approach first performs preliminary human detection to address the camera perspective variance issue. Then, the preliminary results are further refined by information from the framework's temporal layers to jointly

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recognize the detected pedestrians' current status. Therefore, it uses minimal resources to detect abnormality directly on remote drones. We deployed the live system at a remote field with real-time tracking and status reports delivered to the user through wireless transmission (Fig. 1) and provided comprehensive numerical results on the model performance.

II. Related Works

Recent advancements in deep learning provide a robust framework for developing applications for vision-based problems. In contrast to traditional computer vision techniques with handcrafted features, deep neural networks can learn features directly from training data as a data-driven approach, achieving superior results than previous approaches.

A. Pedestrian Detection

Recent works on convolutional and recurrent neural networks enable deformation and occlusion handing [9], using a convolutional neural network (CNN) to detect objects of interest from UAV imagery [10], and using multiple part detectors to combine as a pedestrian detector [11]. Object detections mainly utilize a sliding window approach or object proposal mechanism [12]. However, they do not jointly reason the detected object with their temporal information, relying only on convolutional layers. Recent works have framed object detection as a regression problem [13] by densely generating bounding boxes with spatial separation afterward.

B. Activities Understanding

Action understanding helps provide context on the detected person's current activities. It is also a crucial aspect of a disaster response system as it provides contextual and situational information for the captured scenes [14]. Most current approaches to multipedestrian action recognition take a sequential approach by separating each component and optimizing each part separately. Human is first detected in a CNN model and tracked with an algorithm and then feature representations are extracted for each person to reason on their action [15]. This approach requires much processing time as it needs to repeat the process for each detected person, which does not scale well on a UAV system.

C. Attentive Mechanism

Recent works on the visual attention model gained a lot of empirical achievements, which is particularly important for interpreting the recognition of objects in cluttered scenes [16]. They act as a selection mechanism that encodes the notion of relevance regarding the overall rich stream of visual data. Visual attention can guide the computational models, such as a neural network, to essential parts of the scene to gather more local information [17]. Attention map has been proven to be an effective way to allow neural nets to focus on local details that are more semantically important. However, most approaches require a separate neural network to generate an additional attention layer, which is computationally prohibited on embedded systems with limited computational power.

D. Embedded Systems in UAVs

Most meaningful tasks for autonomous operations of UAVs require a combination of advanced sensors, complex image processing procedures, and flight control algorithms [18]. Hence, traditional algorithms and models that perform well in a ground station might not be suitable in the embedded system in UAVs. For example, [19] discussed the tradeoff between speed, power consumption, and the weight of specific hardware platforms when operating in UAVs. Advanced and computationally expensive algorithms are not directly applicable to operation on embedded devices. Horstrand et al. [20] proposed an application-oriented solution combining UAVs with advanced sensors and GPS, an embedded device, and a hyperspectral camera for monitoring vegetation. Successful usage of vision-based approaches in UAVs includes applications such as Friesian cattle recovery [21], counting vehicles in traffic [22], and drone safe landing [23]. The computation and sensor requirements for UAVs' computer vision tasks are often different than that of an offline counterpart [24]. Therefore, it is necessary to consider the physical hardware limitations when designing models on UAVs.

Algorithm 1 Overall Model Description

while *Not terminated* **do**
  $\mathcal{I}_t \leftarrow$ fetch latest image from camera at t
  $\mathcal{F}_t \leftarrow$ extract multiscale dense features from $\mathcal{I}_t$
  $\mathcal{S}_t, \mathcal{R}_t \leftarrow$ outputs from two separated convolutional layers with $\mathcal{F}_t$ as input
  $\hat{\boldsymbol{b}}_t \leftarrow \texttt{BoxGenerator}(\mathcal{S}_t, \mathcal{R}_t)$ ▷ *Alg.2*
  $\hat{\boldsymbol{f}}_t \leftarrow$ crop $\mathcal{F}_t$ with each $b_t \in \hat{\boldsymbol{b}}_t$
  associate each $f_t \in \hat{\boldsymbol{f}}_t$ with the corresponding $f_{t-1} \in \hat{\boldsymbol{f}}_{t-1}$ from previous time step
  generate $A_{t,i}$ and concat to each $f_t \in \hat{\boldsymbol{f}}_t$ ▷ *Eq.3*
  $\hat{\boldsymbol{a}}_t, \hat{\boldsymbol{c}}_t \leftarrow$ predict actions and confidence scores from the recurrent neural network with input $\hat{\boldsymbol{f}}_t | \hat{\boldsymbol{f}}_{t-1}$
  use $\hat{\boldsymbol{c}}_t$ with non-maximum suppression on $\hat{\boldsymbol{b}}_t$ and $\hat{\boldsymbol{a}}_t$
  output $\hat{\boldsymbol{b}}_t$, $\hat{\boldsymbol{a}}_t$, $\hat{\boldsymbol{c}}_t$ to user

III. Methodology

The overview of our entire architecture is shown in Fig. 2, and overall description is given in Alg. 1. Our architecture is composed of a pedestrian detection model, specifically designed for aerial surveying, as the entry point of the architecture. Each video frame at time t, which we denoted with $\mathcal{I}_t$, is input into the detection network (Section III-A). The output is a dense multiscale feature $\mathcal{F}_t$, of which we will extract the bounding boxes into a fixed size $\hat{\boldsymbol{f}}_t \in \mathbb{R}^{N_t \times D \times M^2}$ representation. The following components further extract temporal information from the dense features, which will be referred to as *ActivityNet* (Section III-B). Finally, the model outputs multiple bounding boxes on pedestrians detection with associated multidimensional predicted actions and confidence scores.

A. POINet

This section will formulate the pipeline and architecture of our Position- and Orientation-Invariant detection Network (POINet) model. This model is responsible for producing bounding boxes of any detected pedestrian in our framework. The challenge lies in the UAVs' operating camera angles, which often vary significantly due to the problem setup's nature. We incorporated traditional computer vision in the generation of bounding boxes. The use of fast and robust

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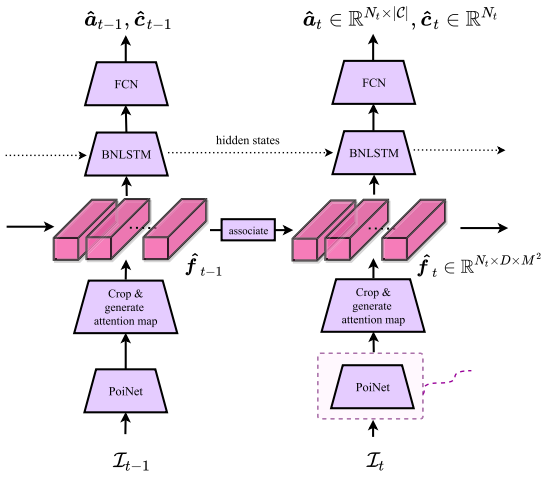


Fig. 2. Overview of our entire framework. Frame $\mathcal{I}_t$ are queried at each time step t and passed through our POINet (Fig. 3) to generate bounding boxes $\hat{\boldsymbol{b}}_t$ on the given frame, which is then used to crop on the original dense feature map to extract fixed size $\hat{\boldsymbol{f}}_t$. The extracted dense features are first concatenated with the generated attention map and then passed to a BNLSTM to reason on the temporal information. The final outputs are the probability of each recognized activity $\hat{\boldsymbol{a}}_t$ and confidence scores $\hat{\boldsymbol{c}}_t$ to suppress low-confidence bounding box.

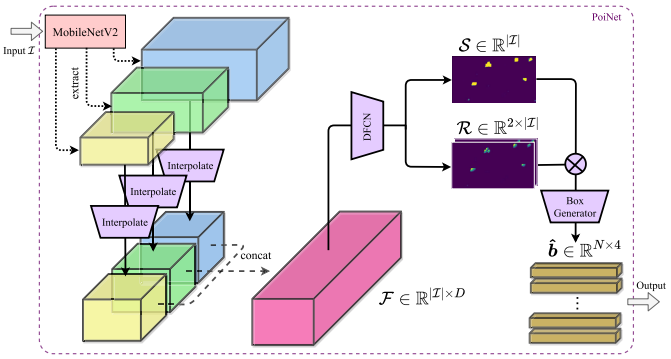


Fig. 3. Overview of the POINetarchitecture, detailing the transformations of the input image data for robust pedestrians detection.

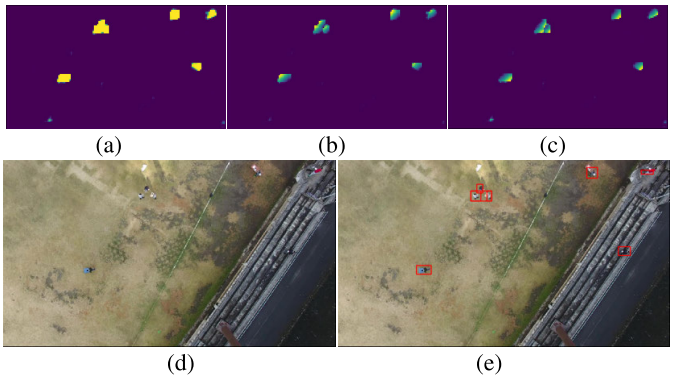


Fig. 4. Generating bounding boxes by extracting spatial information from the images. (a) Segmentation map. (b) Regression map 1. (c) Regression map 2. (d) Input video frame. (e) Output bounding boxes.

computer vision techniques is combined with the extracted features from the CNN for its representation power. Therefore, our approach uses less computational resources and time compared to approaches that use dense proposals of bounding boxes [13]. An example of typical UAV imagery is shown in Fig. 4(e), where the size of pedestrians (and bounding boxes) are incredibly tiny compared to the overall frame size. Furthermore, in aerial surveying, most video feeds are recorded in many different possible orientations, which creates severe challenges to robust detection. POINetaddresses these issues by first extracting multiscale features and then using segmentation and regression maps for anchoring the location of each detected pedestrian.

1) Anchoring Bounding Boxes: The extracted frame $\mathcal{I}_t$ from the input video feed is first fed into some lightweight CNN to extract high-dimensional features. To this end, we used MobileNetV2 for feature extractions aimed at embedded hardware by design [8]. During aerial surveying, pedestrians are often captured at different scales with no fixed zoom level. Therefore, we extract high-dimensional features at different scales and then further concatenate them as multiscale dense features $\mathcal{F}_t$. With the extracted $\mathcal{F}_t$, POINetuses basic convolutional layers to generate segmentation map $\mathcal{S} \in \mathbb{R}^{|W\times H|}$ and regression maps $\mathcal{R} \in \mathbb{R}^{|2\times W\times H|}$. Semantically, the segmentation map $\mathcal{S}$ represents the area of interest that the POINetis trained on, whereas the regression map $\mathcal{R}$ contains the spatial information of the bounding boxes. Segmentation map $\mathcal{S}$ and regression maps $\mathcal{R}$ are each given a ground truth that consists of crucial spatial information to reconstruct bounding boxes [25].

We convert the ground-truth pedestrian location into dense maps $\mathcal{S}$ and $\mathcal{R}$ with the following. For each captured image $\mathcal{I}_t$ at time t, the ground truth consists of a set of bounding boxes $\hat{\boldsymbol{b}} \in \mathbb{R}^{N_t\times 4}$, where $b_i \in \hat{\boldsymbol{b}}$ consists of the location information $(x_0, y_0, x_1, y_1)_t$ and N_t is the number of ground-truth boxes for $\mathcal{I}_t$. $\mathcal{S}$ and $\mathcal{R}$ are defined over all the specific location $\boldsymbol{i} = (i_x, i_y) \in \mathcal{I}$, where we will use $\mathcal{S}_i, \mathcal{R}_i$ to denote their values at (i_x, i_y). For the segmentation map $\mathcal{S}$, we define

$$\mathcal{S}_i = \begin{cases} 1, & \text{if } x_0 \le i_x \le x_1 \wedge y_0 \le i_y \le y_1 \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

such that it overlays areas with pedestrians. Fig. 4(a) demonstrates an example of a segmentation map $\mathcal{S}$, which allows the model to learn the semantic patterns of pedestrians. For the regression map $\mathcal{R}$, each location (i_x, i_y) is a vector $\mathcal{R}_i = [r_0\ r_1]^\top$ that encodes the spatial location—top-left and bottom-right diagonal corners—of the bounding box, respectively. We define $\mathcal{R}_i = [0\ 0]^\top$ if $\mathcal{S}_i = 0$. For pixels i within the segmented area (i.e., $\mathcal{S}_i = 1$), we define $\mathcal{R}_i$ to be in the range of 0 to 1, such that

$$\mathcal{R}_i = \begin{bmatrix} r_0 \\ r_1 \end{bmatrix} = \frac{1}{\alpha} \begin{bmatrix} x_1 - i_x & y_1 - i_y \\ i_x - x_0 & i_y - y_0 \end{bmatrix} \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \tag{2}$$

where $\theta = \arctan(y_1 - y_0)\,/\,(x_1 - x_0)$ is the angle of the diagonal and $\alpha = ((y_1 - y_0)^2 + (x_1 - x_0)^2)^{1/2}$ is the normalizing constant. Fig. 4(b) and (c) illustrates an example of the regression maps $\mathcal{R}$ which allow the model to spatially learn the anchor points for the bounding boxes of pedestrians.

During inference, we begin by first extracting the area of interest by obtaining $\mathcal{R}'$ in Alg. 2 line 2. We will then remove noises in $\mathcal{R}'$ that only contain a tiny contiguous area of patches. Afterward, we apply a maximum filter across $\mathcal{R}'$ [Fig. 4(b) and (c)] to obtain box corner coordinates, which are then used to densely generate multiple bounding boxes $\hat{\boldsymbol{b}}$.

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Algorithm 2 Bounding Box Generator

Parameter : δ: minimum percentage of segmented area
1 **function** BoxGenerator($\mathcal{S}, \mathcal{R}$)
2 $\mathcal{R}' \leftarrow \mathcal{S} \cdot \mathcal{R}$ ▷ *extracts area of interest*
3 remove noise (tiny area of patches) in $\mathcal{R}'$
4 $\hat{\boldsymbol{p}} \leftarrow$ find local maxima in $\mathcal{R}'$ via maximum filter
▷ *now $\hat{\boldsymbol{p}}_1$ contains the top left and $\hat{\boldsymbol{p}}_2$ contains the bottom right coordinates of the candidate boxes*
5 $\hat{\boldsymbol{b}} \leftarrow$ combinations of $\hat{\boldsymbol{p}}_1$ and $\hat{\boldsymbol{p}}_2$
▷ *$\hat{\boldsymbol{b}}$ is a set of box corner coordinates*
6 $\hat{\boldsymbol{b}}' \leftarrow \emptyset$
7 **foreach** $b \in \hat{\boldsymbol{b}}$ **do**
8 $B \leftarrow$ set of pixels enclosed by the box b
9 $B' \leftarrow \left\{ \boldsymbol{i} \in B \,\middle|\, \begin{array}{l} \boldsymbol{i} \text{ is segmented as pixels} \\ \text{of interest in } \mathcal{S} \end{array} \right\}$
10 **if** $|B'| \,/\, |B| \geq \delta$ **then**
11 $\hat{\boldsymbol{b}}' \leftarrow \{b\} \cup \hat{\boldsymbol{b}}'$
12 **return** $\hat{\boldsymbol{b}}'$ ▷ *set of bounding boxes*

Finally, we programmatically remove boxes that contain less than $\delta \in \mathbb{R}, 0 < \delta \leq 1$ fraction amount of segmented pixels (lines 8 to 11), and practically we set $\delta = 0.9$.

2) Incorporating Temporal Information in Pedestrian Detection: While spatial information is crucial for pedestrian detection, temporary changes can often negatively influence the detection accuracy due to changes in perspective or occlusion [26]. Therefore, POINetis designed to densely generate bounding boxes and then jointly refined by the leaked temporal information from the activity layers. *ActivityNet* (Section III-B) is located at the end of our pipeline (Fig. 2), which uses temporal information to predict the current activity of each detected pedestrian and associated confidence scores $\hat{\boldsymbol{c}}_t \in \mathbb{R}^{N_t}$. The confidence scores are then used to jointly refine the bounding boxes $\hat{\boldsymbol{b}} \in \mathbb{R}^{N_t \times 4}$ with nonmaximum suppression [27]. Nonmaximum suppression is a technique to select one entity out of many overlapping entities based on the boxes' probability and the measure of the overlapping region with the Intersection over Union (IoU) metric. We can choose the selection criteria to arrive at the desired results. The leaked confidence scores from the activity component are used to help jointly enhance the refinement of the bounding box proposer.

B. ActivityNet

The activity detection component of our *ActivitytNet* framework takes input directly from POINetto learn temporal information. With each detected pedestrian, a novel attention map-based approach is first used to improve the effectiveness of learning activities (Section III-B1). Afterward, we pass the cropped dense pedestrian features to a Batch Normalized LSTM block [28] as detailed in Section III-B2.

1) Attention Map: We employ a novel attention map-based technique [17] to facilitate adding additional contextual information to each bounding box. It enables a programmatic way to incorporate the bounding boxes' surrounding environment while maintaining the spatial information of the boxes. With the approach in [25] of directly using the cropped boxes as the input features, the boxes will lose much contextual information in the background, which is arguably essential for learning the activities. Here, we are employing a simple, yet effective approach to incorporate more contextual information and the bounding box. In addition, the attention map approach also allows us to avoid introducing distortion of aspect ratio for the cropped images, which is common in many RNN approaches. However, previous studies found that distorted aspect ratio region tends to give higher false-positive bounding boxes [29], where preserving the shape of objects is highly beneficial for accurate classification [30].

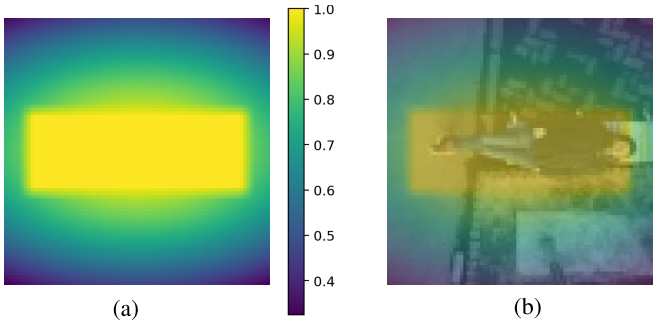


Fig. 5. Attention maps are generated to denote which portion of pixels is more significant for the widened bounding boxes. The yellow rectangular shape in each example can be regarded as the original bounding box. (a) Example of an attention map. (b) Corresponding pedestrian of (a).

The attention matrix A uses a flexible multivariate Gaussian kernel to encode the likely correlation of the expanded region of the bounding box while allowing our temporal network to reason on previously absent contextual information. We define the attention matrix as $A \in \mathbb{R}^{M \times M}$, where $M = \lceil \alpha \cdot \max(W, H) \rceil$ is the dimension of the expanded bounding box, and $\alpha \in \mathbb{R}, \alpha \geq 1$ is a factor that controls the ratio of expansion. The values of the attention matrix are defined as

$$A_i = \begin{cases} 1, & \text{if } x_0 \leq i_x \leq x_1 \wedge y_0 \leq i_y \leq y_1 \\ \varphi(\boldsymbol{i} \mid \boldsymbol{b}_c, \boldsymbol{\Sigma}), & \text{otherwise} \end{cases} \tag{3}$$

where $\varphi(\boldsymbol{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \exp\{-(1/2)(\boldsymbol{x} - \boldsymbol{\mu})\boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\}$ resembles the probability density distribution of the multivariate Gaussian [31] without a normalizing factor. Moreover, $\boldsymbol{\Sigma}$ is a positive definite covariance matrix that encodes our belief on the spatial correlation between the action of the pedestrians and their surrounding environment. Section IV provides experimental results on the effectiveness of the additional attention map. The flexibility of this novel attention map allows us to express the relevance of each pixel to the surrounding contextual information. We formulate an algorithmic way to generate a pseudo-attention map based purely on the given bounding box. In addition, this novel method directly avoids distortion of cropped images, and experimentally, we show that the additions of the attention map enhance the overall model accuracy.

2) Training POINet and ActivityNet End-to-End: We have described a way to obtain a set of reliable detection from raw images in Section III-A. However, temporal information was shown to be important in action recognition and provide a valuable basis for reasoning on detection. Therefore, after obtaining predictions by the BNLSTM component in Fig. 2, we utilize the temporal confidence score $\hat{\boldsymbol{c}}_t$ by jointly refining the detected bounding boxes. That is, we use confidence scores to refine and suppress low-confidence bounding boxes with nonmaximum suppression [27]. The reason is that temporal

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information can often provide a better estimate of whether the captured bounding box contains a pedestrian.

For each frame $\mathcal{I}_t$, given that we had extracted N bounding boxes of pedestrian $\hat{\boldsymbol{b}}_t = \{\boldsymbol{b}_t^1, \ldots, \boldsymbol{b}_t^N\}$, we extract a fixed-sized representation $\hat{\boldsymbol{f}}_t = \{f_t^1, \ldots, f_t^N\}$ for each box from the multiscale dense feature $\mathcal{F}_t$. These embeddings are then sent to our BNLSTM block, which produces predictions on their corresponding activities $\hat{\boldsymbol{a}}_t$ and confidence scores $\hat{\boldsymbol{c}}_t$ that denote the likelihood that the sequence of frames contains actual pedestrians. Finally, we construct a loss that accounts for the multiactivities nature of the dataset. Each pedestrian can be executing a *primary* and a *secondary* action simultaneously. For example, a pedestrian can be *walking* while *pushing* something, or *sitting* while *reading* a book. Let $\bar{\boldsymbol{p}}_{t,p}^i$, $\bar{\boldsymbol{p}}_{t,s}^i$ for $i \in \{1, \ldots, N_t\}$ be the corresponding one-hot-encoded ground truth of *primary* and *secondary* activities, where N_t is the number of bounding box at frame t. We will match our predictions using the closest bounding box distance, which gives us the prediction vector $\boldsymbol{p}_{t,p}^i$ and $\boldsymbol{p}_{t,s}^i$ for $i \in \{1, \ldots, N_t\}$. We can now express the loss function as

$$\mathcal{L}_{ps} = -\frac{1}{T}\sum_{t=1}^{T}\left(\frac{1}{N_t \cdot N_p}\sum_{i=1}^{N_t}\bar{\boldsymbol{p}}_{t,p}^i \log \boldsymbol{p}_{t,p}^i - \epsilon_w \frac{1}{N_t \cdot N_s}\sum_{i=1}^{N_t}\bar{\boldsymbol{p}}_{t,s}^i \log \boldsymbol{p}_{t,s}^i\right) \quad (4)$$

where T is the total number of frames, N_p and N_s are the number of labels for primary and secondary actions, respectively, and ϵ_w is a weight factor that allows us to balance the two activities differently. In practice, we set $\epsilon_w = 0.5$ to focus more on the *primary* action.

IV. Experimental Results

In this section, numerical results are reported on pedestrian detection and multiaction recognition tasks, comparing multiple baselines. We also provide empirical results on the deployment of our networks on an embedded device. In particular, the pipeline architecture is deployed on live UAVs to test its field performance and transmission speed. The performance and practicality of the system are evaluated in this section. We used an IoU of 0.5 as the threshold for the following results.

A. Dataset Description

We evaluate our architecture on the *Okutama-Action* dataset [32]. It is a publicly available multipedestrian drone dataset that contains labels at each frame that describe the pedestrian's current activities. Deploying a reliable drone-based response system requires recognizing the pedestrians' active state to detect their distress levels. The dataset consists of 77 K fully-annotated frames, where the video sequences are in 4 K resolution and 30 frames/s. Multiple pedestrians exist in a given frame, with labels describing the pedestrians' current activities. The frames contain varying altitudes, ranging from about 10 to 45 m, and varying camera angles from 45° to 90° (bird's eye view). The current dataset consists of a labeled bounding box around each person and one or more person's actions. For example, a person can perform one *primary* action, such as standing or walking, and one *secondary* action, such as carrying or pulling.

TABLE I
Multiaction Recognition Performed on the Embedded System TX2. Inference Time Is in ($\mu \pm 2\sigma$) Seconds

Method	mAP (%) (*primary* / *secondary*)	Inf. time (sec)
Ours	24.2 / 13.6	1.43 ± 0.21
Ours (w/o att.)	22.5 / 12.3	1.41 ± 0.23
SSD RGB [33]	18.8 / 7.41	2.02 ± 0.39
SSD Optical Flow [33]	6.47 / 2.21	1.91 ± 0.37
SSD-MobileNetV2 [8]	17.2 / 7.10	1.88 ± 0.22
Inception-SSD [34]	19.6 / 9.51	2.02 ± 0.39
YOLOv3-LSTM [35]	20.2 / 9.96	2.21 ± 0.65
RAER [36]	21.5 / 10.56	3.16 ± 0.64

B. Models for Multiactivities Prediction

We evaluate against 1) SSD RGB [33]: pretrained on ImageNet and fine-tuned to predict individual actions based on fixed-sized images of individual pedestrians; 2) SSD Optical Flow [32]: a variant of the previous baseline that uses motion cues from the optical flow steam; 3) SSD-MobileNetV2 [8]: similar to SSD RGB baseline, but instead MobileNet is used as the feature extractor as a lightweight variant; 4) Inception-SSD [34]: similar to the previous baseline, but instead Inception block is added as an extra layer in the SSD before the prediction; 5) YOLOv3 [35]: pretrained and fine-tuned to the target drone dataset, with an additional LSTM block at the end; 6) RAER [36]: a hierarchical autoencoder-based model that combines scene information with individual action. We follow the main proposed architecture of four relational layers and focus on the whole scene.

C. Models for Pedestrian Detection

we evaluate against 1) ReInspect [37]: a joint multiperson detection designed for the crowded scene; 2) ReInspect-rezoom [37]: similar to the last baseline, but with an additional re-zooming layer that transforms features into a scale-invariant representation; 3) Faster-RCNN [12]: a widely adopted region proposal model; 4) SSD [38]: similar to the model that we evaluate in activities prediction, but focus only on proposing bounding boxes; 5) YOLOv3 [33]: similarly focuses on bounding boxes proposal.

Implementation Details: All of our models are trained using the same optimization and backpropagation scheme. We use the Adam stochastic gradient descent optimizer, with an initial learning rate set to 10^{-4} and fixed hyperparameters to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. We begin by training the pedestrian detection component of each network, which involves using the given ground-truth box coordinates to propose bounding boxes. After a fixed number of epochs, we jointly train the temporal components. Gradients are allowed to pass through the previous part, which facilitates each component to compensate for the other in an end-to-end setting.

D. Model Deployment on Embedded-UAV

The overall drone and the onboard system are illustrated in Fig. 6. The coding environment is developed on top of

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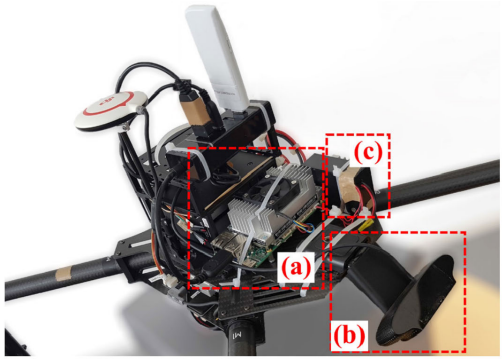


Fig. 6. DJI Matrice 100 drone with an embedded TX2 device used for the field test and inference benchmark. (a) NVIDIA Jetson TX2, (b) Logitech C920 camera, and (c) voltage regulator.

the Python Deep-Learning *PyTorch* library. Then, the trained model is converted to a Caffe model that can be run natively without the Python runtime environment. We packaged the model as an onboard application and deployed it to the NVIDIA Jetson TX2. The NVIDIA Jetson TX2 is an embedded AI computing device that enables edge computing with an NVIDIA Pascal GPU with 8 GB RAM. The Jetson TX2 then acted as the onboard computer and attached to the DJI Matrice 100 [Fig. 6(a)]. The TX2 is mounted on the Auvidea J120 carrier board and connects to the UAV using a USB to TTL/UART connection. We attached a Logitech C920 camera [Fig. 6(b)] to capture video footage, and an LTE dongle to the Jetson TX2 to communicate information back to the user. The onboard battery for the UAV is a DJI TB48D battery with 130 Wh, acting as the power source to the onboard computer through a voltage regulator [Fig. 6(c)]. The total payload of the entire onboard setup during the field test (Fig. 1) is at around 0.57 kg.

During deployment, an image frame is first fetched from the video feed, resized, and sent to the Caffe model. The frame extraction procedure is performed using the OpenCV C++ library, typically taking < 5 ms. Then, the frame is passed to the POINet, followed by the ActivityNet for the whole inference procedure. The embedded processing device typically takes around 2–3 s to process each frame and transmit the corresponding information. Depending on the number of detected pedestrians, the drone will then uses around 100–500 bytes of network bandwidth to send the information back to the remote operator.

E. Multipedestrian Multiactivities Recognition

In Table I, we report the mean average precision (mAP) and inference time on the NVIDIA Jetson TX2 embedded on UAV during deployment. We report accuracies separately for the primary and secondary actions by the pedestrians. We take the top two predicted outputs as primary and secondary actions for models that do not predict multiple actions. We also report a variant of our model without the additional attention map (Section III-B1) to evaluate its effectiveness. Inference time performed on the embedded device reports the mean and the 95% quantile over 30 runs in seconds.

Overall, our model achieves state-of-the-art performance for both *primary* and *secondary* actions recognition even without the ground-truth locations of the individuals. The comparison to SSD variants, which propose pedestrians' action based purely on captured frames, indicates that having a temporal joint representation shared across multiple tasks helps us improve accuracy. YOLOv3-LSTM and RAER perhaps are the main competitor as they utilize temporal information across frames. Of the two, RAER performs better as it utilizes a hierarchical structure and scene information. However, the architecture also imposes more overhead in inference time, which is essential in a real-time UAV application with limited processing power. In fact, our model requires the lowest inference time as our model requires fewer parameters and uses operations that cut down runtime. Our attention map approach also helps to improve performance, which is likely due to the extra information around the boundary that would otherwise be cropped (Fig. 5). Such an approach also helps to maintain the aspect ratio of the input to our temporal component. Our algorithmic attention map will likely be less performant than a learning approach; however, it requires little to no overhead to generate, which is essential in time-sensitive applications. Figs. 1 and 4 are results from our actual running model.

TABLE II

RESULTS FOR MULTIPEDESTRIAN DETECTION FROM UAV IMAGERY. INFERENCE TIME IS IN $(\mu \pm 2\sigma)$ SECONDS

Method		mAP (%)	Inf. time (sec)
POINet	(ours)	84.5	1.52 ± 0.16
ReInspect	[37]	79.8	3.41 ± 0.77
ReInspect-rezoom	[37]	81.1	3.50 ± 0.69
Faster-RCNN	[12]	68.1	2.98 ± 0.42
SSD	[33]	72.3	1.98 ± 0.37
YOLOv3	[35]	77.2	2.38 ± 0.52

F. Multipedestrian Detection

In addition to evaluating the overall performance of action recognition, we also conducted a standalone experiment evaluating the performance for POINetto propose bounding boxes from UAV imagery. Our evaluation of pedestrian detection from the UAV dataset (Table II) found that existing box proposal approaches do not work well in sparse scenes with varying camera angles. For example, our leading contender ReInspect works well in a dense proposal of a crowded scene as shown in [37]. However, it does not perform well in UAV imagery, mainly when the objects of interest are sparsely spaced, which introduces more false positives in the model output. The addition of an invariant layer in ReInspect-rezoom helps enhance the performance as it scales better with smaller objects. However, both models require quite a significant runtime overhead due to the need to make joint predictions in a sequential manner. It should be noted that SSD, YOLO, and RCNN models are built to propose bounding boxes for various objects. That is, they are capable of proposing bounding boxes with more than one class as they are designed for dense proposals of day-to-day objects. However, for a domain-specific application where we are only interested in search and rescue in a sparse environment (e.g., in a wide open field), these model assumptions do not seem to perform well.

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Therefore, using them solely for pedestrian detection seems to limit their predictability in our UAV application. By exploiting the domain specificity, our proposed POINetachieves higher precision with less computational overhead.

V. Conclusion

We have proposed a novel approach for real-time UAV applications by exploiting domain knowledge. Our model is designed to excel in detecting pedestrians regardless of their orientation from the UAV's perspective. It is lightweight and combines traditional computer vision techniques to acquire the anchor points of boxes: harnessing the rich representation power of the deep-learning model for robust drone operations.

References

[1] S. Schulte, F. Hillen, and T. Prinz, "Analysis of combined UAV-based RGB and thermal remote sensing data: A new approach to crowd monitoring," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 347–354, Jan. 2017. [Online]. Available: https://www.int-arch-photogramm-remote-sens-spatial-inf-sci.net/XLII-2-W6/347/2017/

[2] M. Abdelkader, M. Shaqura, C. G. Claudel, and W. Gueaieb, "A UAV based system for real time flash flood monitoring in desert environments using Lagrangian microsensors," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, May 2013, pp. 25–34.

[3] H. Hildmann and E. Kovacs, "Review: Using unmanned aerial vehicles (UAVs) as mobile sensing platforms (MSPs) for disaster response, civil security and public safety," *Drones*, vol. 3, no. 3, p. 59, Jul. 2019.

[4] M. Aljehani and M. Inoue, "Multi-UAV tracking and scanning systems in M2M communication for disaster response," in *Proc. IEEE 5th Global Conf. Consum. Electron.*, Oct. 2016, pp. 1–2.

[5] Y. Karaca et al., "The potential use of unmanned aircraft systems (drones) in mountain search and rescue operations," *Amer. J. Emergency Med.*, vol. 36, no. 4, pp. 583–588, 2018.

[6] Y. M. Chen, L. Dong, and J.-S. Oh, "Real-time video relay for UAV traffic surveillance systems through available communication networks," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Mar. 2007, pp. 2608–2612.

[7] Y. Bai, W. Du, Z. Ma, C. Shen, Y. Zhou, and B. Chen, "Emergency communication system by heterogeneous wireless networking," in *Proc. IEEE Int. Conf. Wireless Commun., Netw. Inf. Secur.*, Jun. 2010, pp. 488–492.

[8] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[9] W. Ouyang and X. Wang, "Joint deep learning for pedestrian detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 2056–2063.

[10] M. B. Bejiga, A. Zeggada, A. Nouffidj, and F. Melgani, "A convolutional neural network approach for assisting avalanche search and rescue operations with UAV imagery," *Remote Sens.*, vol. 9, no. 2, p. 100, 2017.

[11] Y. Tian, P. Luo, X. Wang, and X. Tang, "Deep learning strong parts for pedestrian detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1904–1912.

[12] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[13] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[14] J. Yang, X. Xie, and W. Yang, "Effective contexts for UAV vehicle detection," *IEEE Access*, vol. 7, pp. 85042–85054, 2019.

[15] V. Ramanathan, J. Huang, S. Abu-El-Haija, A. Gorban, K. Murphy, and L. Fei-Fei, "Detecting events and key actors in multi-person videos," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3043–3053.

[16] A. Borji and L. Itti, "State-of-the-art in visual attention modeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 185–207, Jan. 2013.

[17] H. Zheng, J. Fu, T. Mei, and J. Luo, "Learning multi-attention convolutional neural network for fine-grained image recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 5209–5217.

[18] A. Al-Kaff, D. Martín, F. García, A. De La Escalera, and J. M. Armingol, "Survey of computer vision algorithms and applications for unmanned aerial vehicles," *Exp. Syst. Appl.*, vol. 92, pp. 447–463, Feb. 2017.

[19] D. Hulens, J. Verbeke, and T. Goedemé, "How to choose the best embedded processing platform for on-board UAV image processing?" in *Proc. 10th Int. Conf. Comput. Vis. Theory Appl.*, 2015, pp. 1–10.

[20] P. Horstrand, R. Guerra, A. Rodriguez, M. Diaz, S. Lopez, and J. F. Lopez, "A UAV platform based on a hyperspectral sensor for image capturing and on-board processing," *IEEE Access*, vol. 7, pp. 66919–66938, 2019.

[21] W. Andrew, C. Greatwood, and T. Burghardt, "Aerial animal biometrics: Individual friesian cattle recovery and visual identification via an autonomous UAV with onboard deep inference," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Nov. 2019, pp. 237–243.

[22] G. Amato, L. Ciampi, F. Falchi, and C. Gennaro, "Counting vehicles with deep learning in onboard UAV imagery," in *Proc. IEEE Symp. Comput. Commun. (ISCC)*, Jun. 2019, pp. 1–6.

[23] E. Kakaletsis et al., "Computer vision for autonomous UAV flight safety: An overview and a vision-based safe landing pipeline example," *ACM Comput. Surv.*, vol. 54, no. 9, pp. 1–37, 2021.

[24] A. Douklias, L. Karagiannidis, F. Misichroni, and A. Amditis, "Design and implementation of a UAV-based airborne computing platform for computer vision and machine learning applications," *Sensors*, vol. 22, no. 5, p. 2049, Mar. 2022.

[25] T. Bagautdinov, A. Alahi, F. Fleuret, P. Fua, and S. Savarese, "Social scene understanding: End-to-end multi-person action localization and collective activity recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4315–4324.

[26] X. Wang, T. X. Han, and S. Yan, "An HOG-LBP human detector with partial occlusion handling," in *Proc. IEEE 12th Int. Conf. Comput. Vis.*, Sep. 2009, pp. 32–39.

[27] A. Neubeck and L. Van Gool, "Efficient non-maximum suppression," in *Proc. 18th Int. Conf. Pattern Recognit. (ICPR)*, 2006, pp. 850–855.

[28] T. Cooijmans, N. Ballas, C. Laurent, C. Gulcehre, and A. C. Courville, "Recurrent batch normalization," in *Proc. 5th Int. Conf. Learn. Represent.*, 2017, pp. 1–13.

[29] Z. Xu, X. Xin, L. Wang, R. Yang, and F. Pu, "Deformable ConvNet with aspect ratio constrained NMS for object detection in remote sensing imagery," *Remote Sens.*, vol. 9, no. 12, p. 1312, Dec. 2017.

[30] L. Zheng, Y. Zhao, S. Wang, J. Wang, and Q. Tian, "Good practice in CNN feature transfer," 2016, *arXiv:1604.00133*.

[31] C. M. Bishop, *Pattern Recognition and Machine Learning*. Berlin, Germany: Springer, 2006.

[32] M. Barekatain et al., "Okutama-Action: An aerial view video dataset for concurrent human action detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 28–35.

[33] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 21–37.

[34] W. Chen, Y. Qiao, and Y. Li, "Inception-SSD: An improved single shot detector for vehicle detection," *J. Ambient Intell. Humanized Comput.*, vol. 13, no. 11, pp. 5047–5053, Jun. 2020.

[35] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[36] M. S. Ibrahim and G. Mori, "Hierarchical relational networks for group activity recognition and retrieval," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 721–736.

[37] R. Stewart, M. Andriluka, and A. Y. Ng, "End-to-end people detection in crowded scenes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2325–2333.

[38] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.



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